Embodied Interpretability

Linking Causal Understanding to Generalization in Vision-Language-Action Models

Michał Jan Włodarczyk

CCAI Research Seminar.
Monday 10.08.2026r.

Agenda:

- **Preliminaries** - what are VLAs and where are they used?
- **Embodied Interpretability** - the paper that was selected.
- **VLA Interpretability -** what has been done so far?
- **Questions & Discussion**

Preliminaries

- What are VLAs?
- Architecture and components.
- Current model landscape.
- Mathematical formulation.
- Utilization domains.

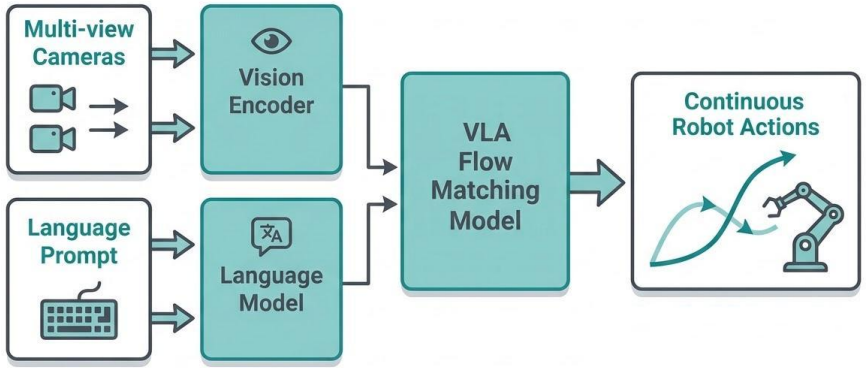
What are Vision-Language-Action models

Learn:

- Policy π_θ parametrized by θ, mapping
  Multimodal observations to actions.

- VLA models map raw camera views directly to multi-joint command vectors,

- Continuous Action Space VLAs predict joint control actions and avoid discretization to minimize serrors.



Visual Representation of the model scheme used Embodied Interpretability Intelligence et al. 2025

Mathematical Formulation.

Formally, the policy predicts an action $\mathbf{a}_t$ at timestep t conditioned on a multimodal context $\mathbf{X}_t$:

$$\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{X}_t)$$

The context $\mathbf{X}_t$ is constructed by concatenating the sequence of visual patch embeddings $(\mathbf{v}_1, \ldots, \mathbf{v}_{n_1}) \in \Omega$ and the prompt instruction tokens $(\mathbf{l}_1, \ldots, \mathbf{l}_{n_2}) \in O$:

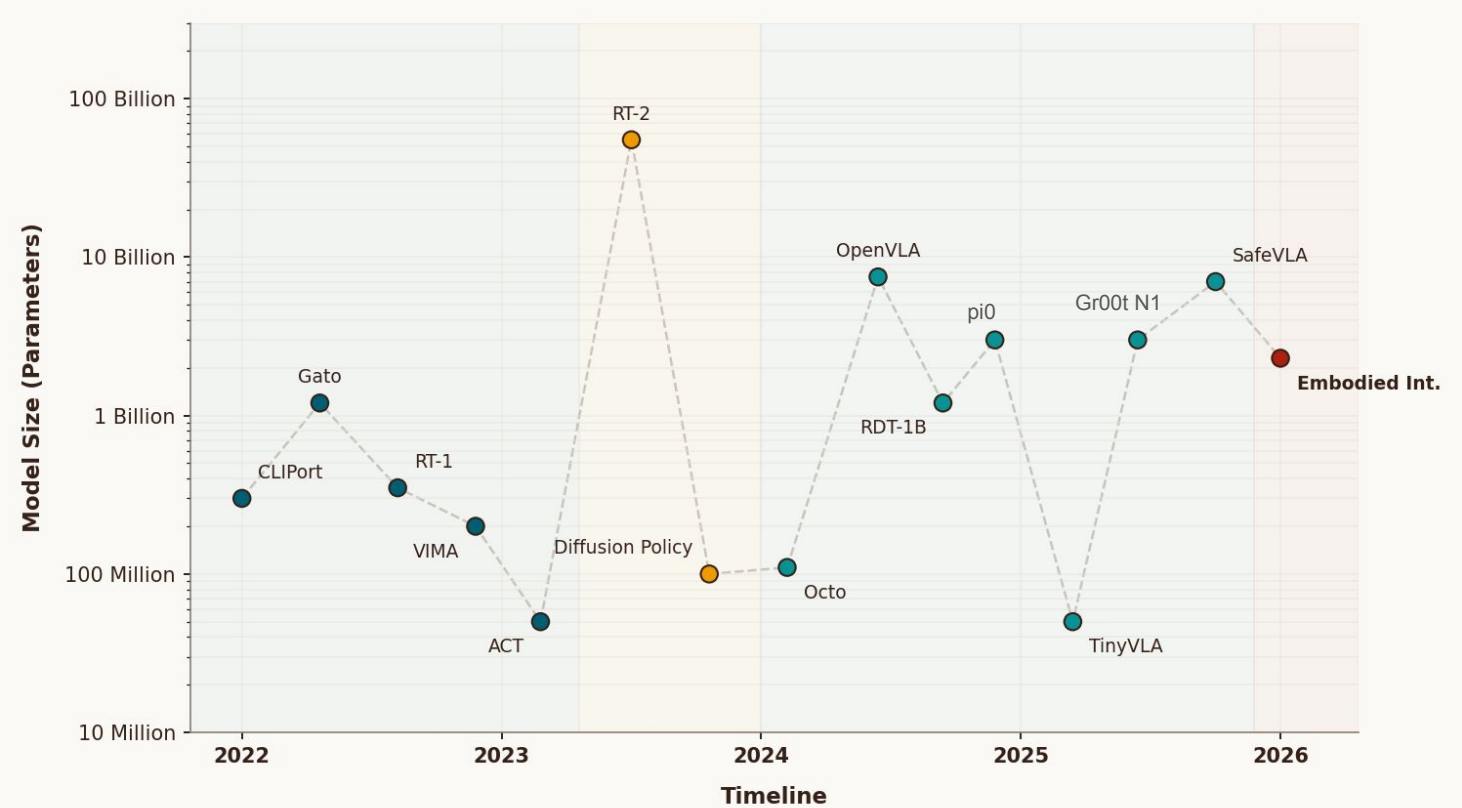
$$\mathbf{X}_t = [\mathbf{v}_1, \ldots, \mathbf{v}_{n_1}, \mathbf{l}_1, \ldots, \mathbf{l}_{n_2}] \in \mathbb{R}^{(n_1+n_2)\times d}$$

Where n_1 and n_2 denote the sequence level lengths of visual patches and instruction tokens and d is the embedding dimension.

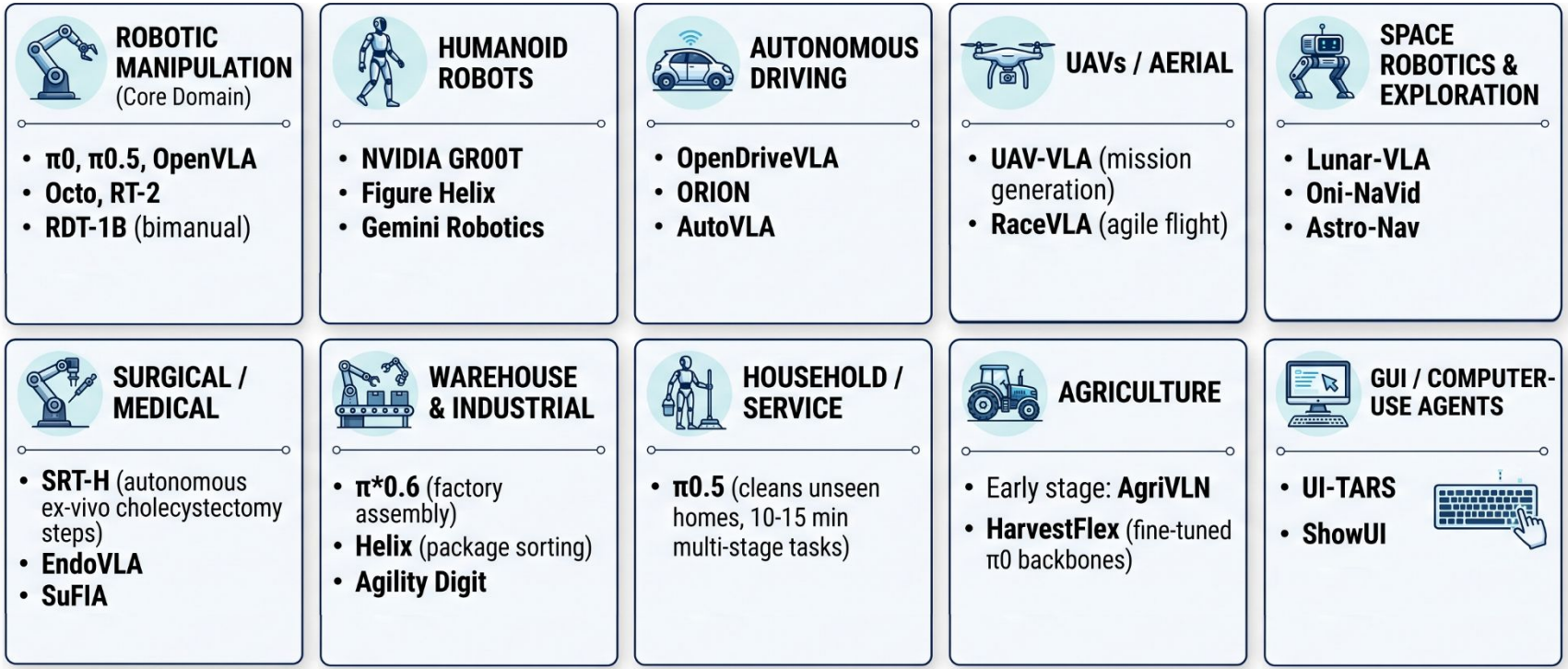
Architecture and components.

Model	Visual encoder	Language model	Trajectory decoder
OpenVLA **7B**	SigLIP + DINOv2, fused per patch. One camera.	Llama-2-7B Image and instruction as one token sequence.	**Discrete** Actions become text: 256 bins per joint, one action per forward pass.
π0 / π0.5 **3.3B**	SigLIP (from PaliGemma). Multiple cameras.	PaliGemma 3B + small 300M action expert reading robot state.	**Continuous** Flow matching denoises 50-step action chunks - no quantization, 50 Hz control.

Current model landscape.



Utilization domains.



Embodied Interpretability

- Paper Info.
- Authors & Affiliations.
- Core contributions.
- Results.
- Reviews & Rebuttal.

Embodied Interpretability

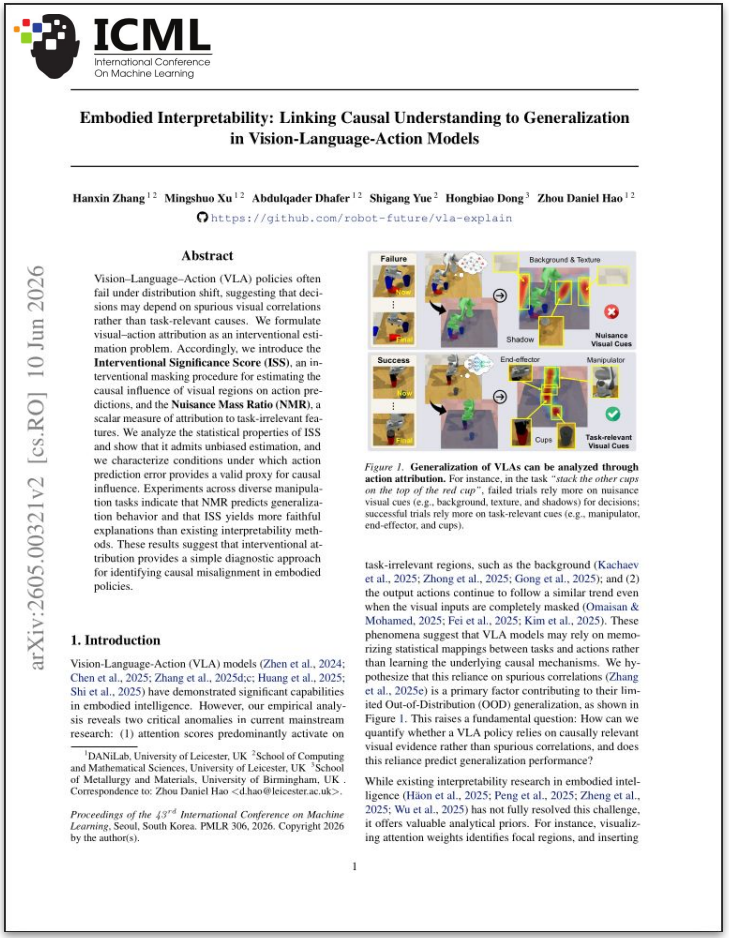
Linking Causal Understanding to Generalization in Vision-Language-Action Models

Problem:

- VLA policies often fail under distribution shift,
- Their decisions may depend on spurious visual correlations rather than task-relevant causes,
- Attention activates on task-irrelevant regions, and actions follow similar trends even with fully masked visual input

Method:

- Frames visual-action attribution as an interventional estimation problem.



ICML International Conference On Machine Learning

Embodied Interpretability: Linking Causal Understanding to Generalization in Vision-Language-Action Models

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https://github.com/robot-future/vla-explain

arXiv:2605.00321v2 [cs.RO] 10 Jun 2026

Abstract

Vision–Language–Action (VLA) policies often fail under distribution shift, suggesting that decisions may depend on spurious visual correlations rather than task-relevant causes. We formulate visual–action attribution as an interventional estimation problem. Accordingly, we introduce the **Interventional Significance Score (ISS)**, an interventional masking procedure for estimating the causal influence of visual regions on action predictions, and the **Nuisance Mass Ratio (NMR)**, a scalar measure of attribution to task-irrelevant features. We analyze the statistical properties of ISS and show that it admits unbiased estimation, and we characterize conditions under which action prediction error provides a valid proxy for causal influence. Experiments across diverse manipulation tasks indicate that NMR predicts generalization behavior and that ISS yields more faithful explanations than existing interpretability methods. These results suggest that interventional attribution provides a simple diagnostic approach for identifying causal misalignment in embodied policies.

Figure 1. **Generalization of VLAs can be analyzed through action attribution.** For instance, in the task *"stack the other cups on the top of the red cup"*, failed trials rely more on nuisance visual cues (e.g., background, texture, and shadows) for decisions; successful trials rely more on task-relevant cues (e.g., manipulator, end-effector, and cups).

1. Introduction

Vision-Language-Action (VLA) models (Zhen et al., 2024; Chen et al., 2025; Zhang et al., 2025d;c; Huang et al., 2025; Shi et al., 2025) have demonstrated significant capabilities in embodied intelligence. However, our empirical analysis reveals two critical anomalies in current mainstream research: (1) attention scores predominantly activate on task-irrelevant regions, such as the background (Kachaev et al., 2025; Zhong et al., 2025; Gong et al., 2025); and (2) the output actions continue to follow a similar trend even when the visual inputs are completely masked (Omaisan & Mohamed, 2025; Fei et al., 2025; Kim et al., 2025). These phenomena suggest that VLA models may rely on memorizing statistical mappings between tasks and actions rather than learning the underlying causal mechanisms. We hypothesize that this reliance on spurious correlations (Zhang et al., 2025e) is a primary factor contributing to their limited Out-of-Distribution (OOD) generalization, as shown in Figure 1. This raises a fundamental question: How can we quantify whether a VLA policy relies on causally relevant visual evidence rather than spurious correlations, and does this reliance predict generalization performance?

While existing interpretability research in embodied intelligence (Häon et al., 2025; Peng et al., 2025; Zheng et al., 2025; Wu et al., 2025) has not fully resolved this challenge, it offers valuable analytical priors. For instance, visualizing attention weights identifies focal regions, and inserting

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Interventional Significance Score

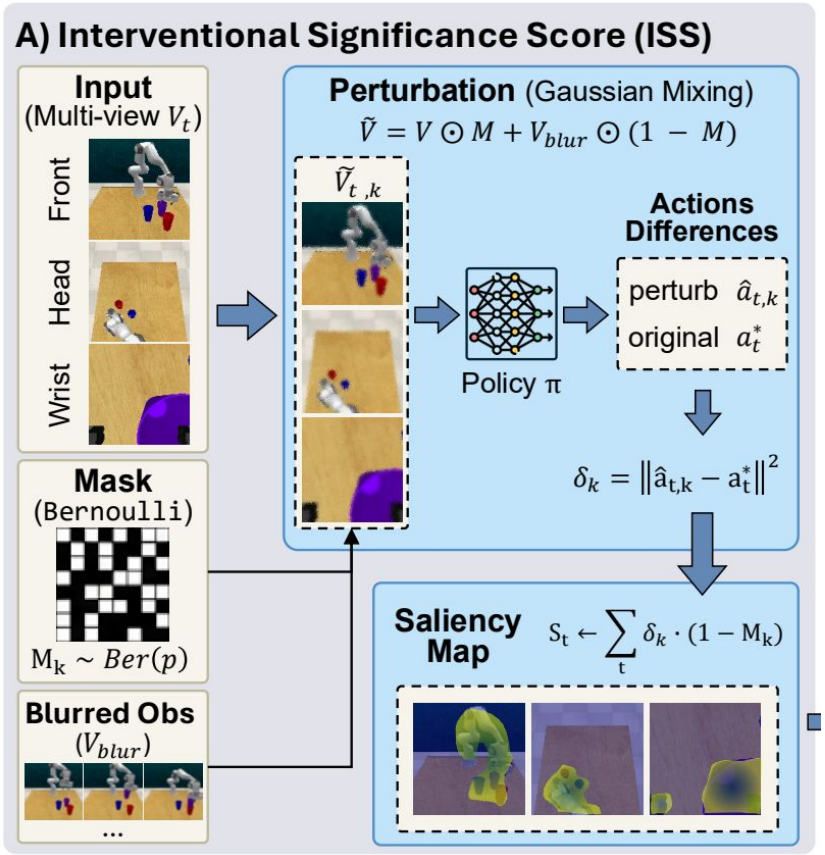
The pipeline as 4 steps:

- Sample random Bernoulli masks.
- Masked-out patches are blurred.
- Run the policy on each perturbed image
- Saliency of a region is how much the action moved ($\|\hat{a} - a^*\|^2$), averaged over masks.

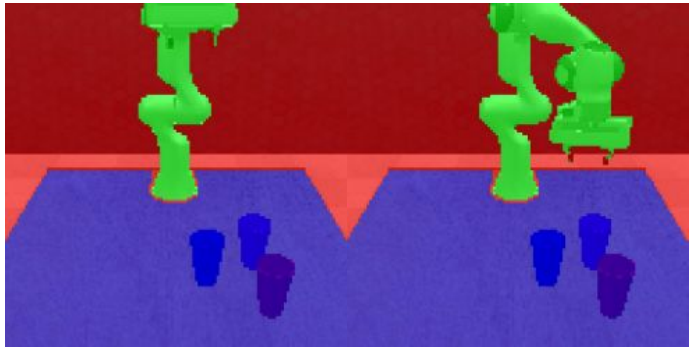
Formal definition:

$$\mathrm{ISS}_i(\theta) = \mathbb{E}_{\mathbf{X}\sim\mathcal{D}}\left[\sum_{t=1}^{T} D_{\mathrm{KL}}\left(\pi_\theta(\cdot \mid \mathbf{X}_t) \,\|\, \pi_\theta(\cdot \mid \tilde{\mathbf{X}}_t^{(i)})\right)\right]$$

Setup note: N=100 masks, p=0.3; ~65× cost of one inference.



NMR - Nuisance Mass Ratio



Taxonomy

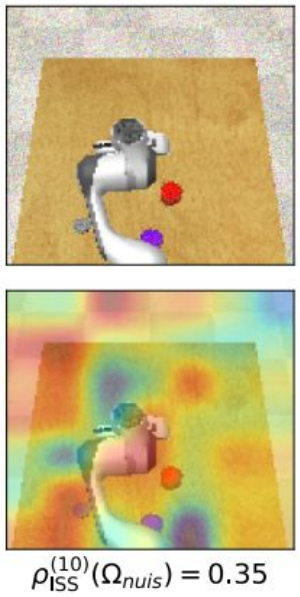
Ω_act - manipulator & end-effector (green)

Ω_sup - task objects, cups & table (blue)

Ω_nuis - irrelevant (walls & shadows) (red)

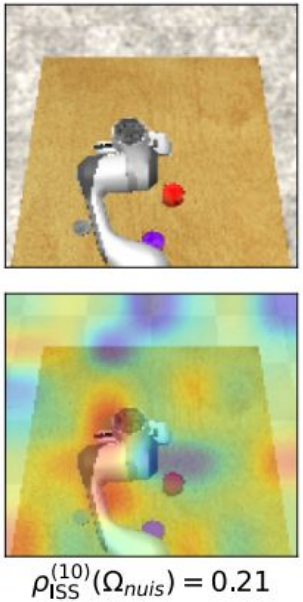
- Expert needs only **Ω_act ∪ Ω_sup** (Causal Markov Blanket of the action)
- **NMR@k** is the top-k% fraction of ISS mass falling in Ω_nuis
- Ideal policy when **NMR ≈ 0**. If it depends on the nuisance regions they call it **"causal hallucination"**

Perturbation Operations



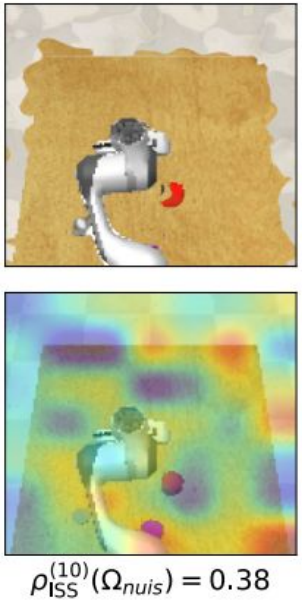
$\rho_{\text{ISS}}^{(10)}(\Omega_{nuis}) = 0.35$

Noise Perturbation



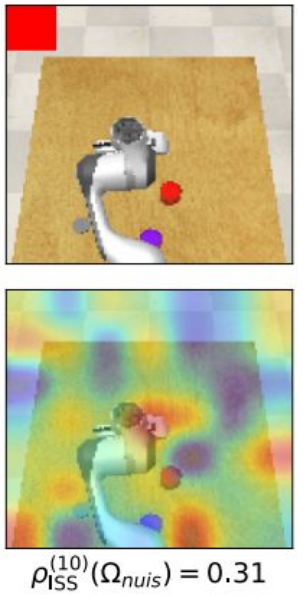
$\rho_{\text{ISS}}^{(10)}(\Omega_{nuis}) = 0.21$

Texture Perturbation



$\rho_{\text{ISS}}^{(10)}(\Omega_{nuis}) = 0.38$

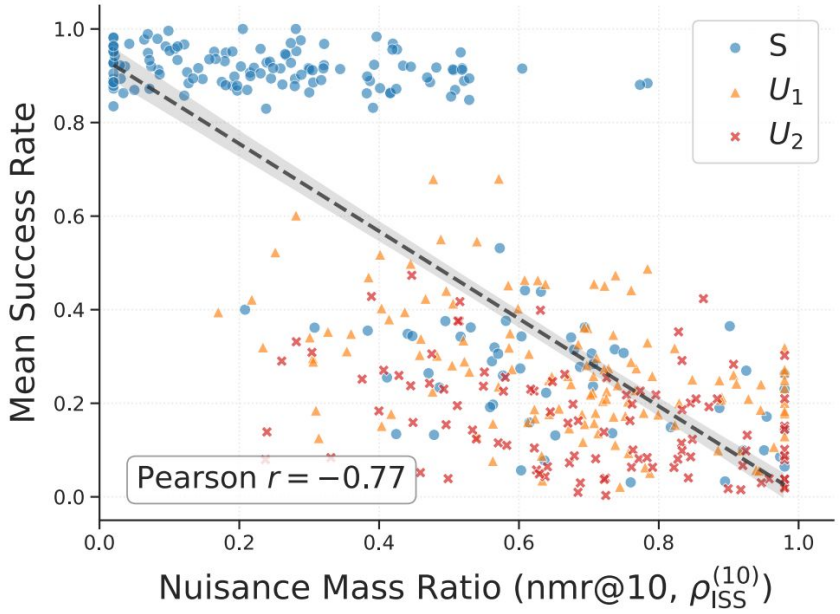
Geometric Perturbation

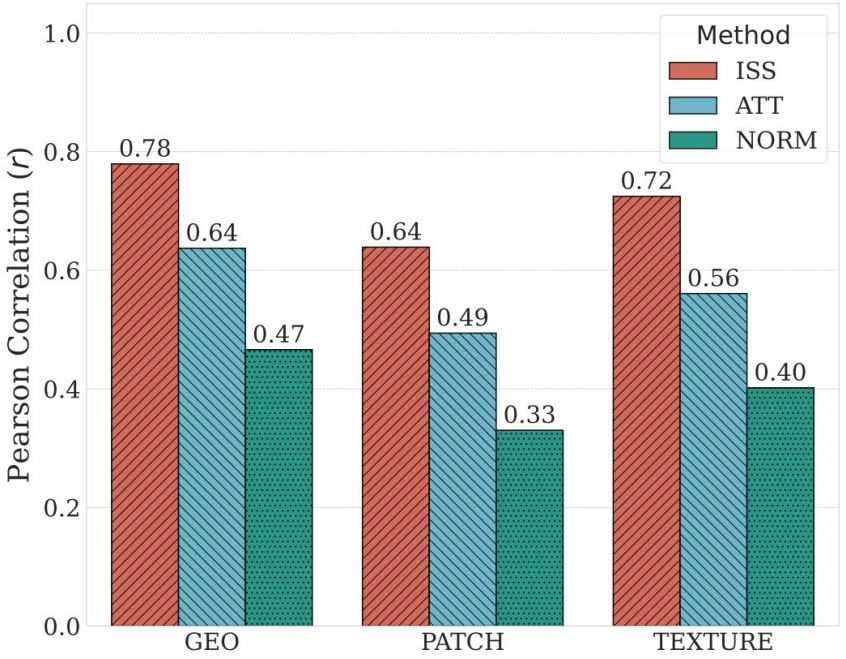


$\rho_{\text{ISS}}^{(10)}(\Omega_{nuis}) = 0.31$

Patch Perturbation

Results & Comparison





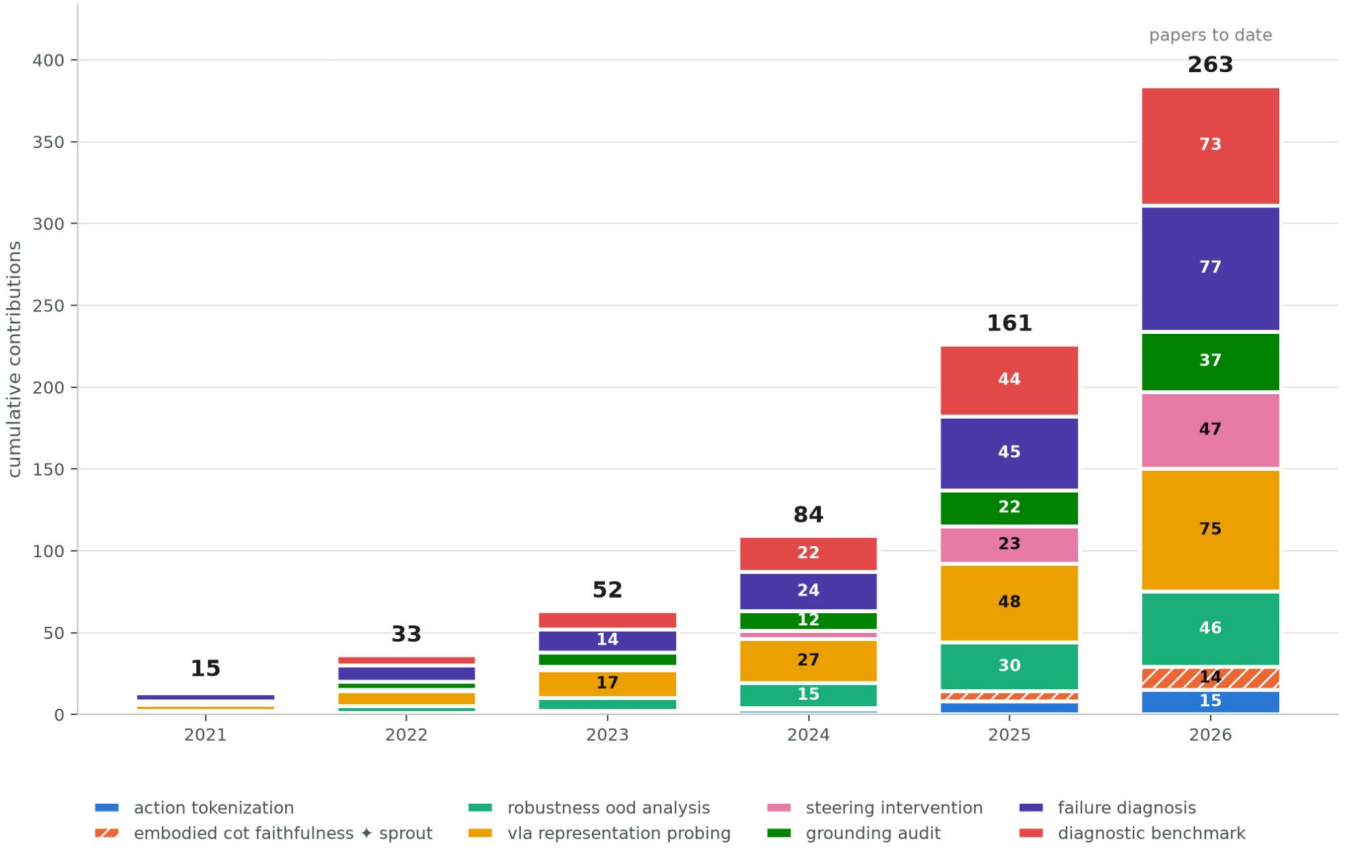
Reviews Main Criticism

- NMR needs ground-truth nuisance masks.
- Unimodal-action assumption is weak and MSE as a KL-divergence proxy may break for other policies.
- Monte Carlo masking is offline and post-hoc only, no real-time monitoring and intervention possible.
- Single architecture, sim-only which means cross-model generality hasn’t been validated.
- Why attribution, not counterfactuals? Clever Hans literature raised.

VLA Interpretability

- Research landscape of VLA Interpretability.
- Well established and saturated topics.
- Current breakthrough topics and research avenues.

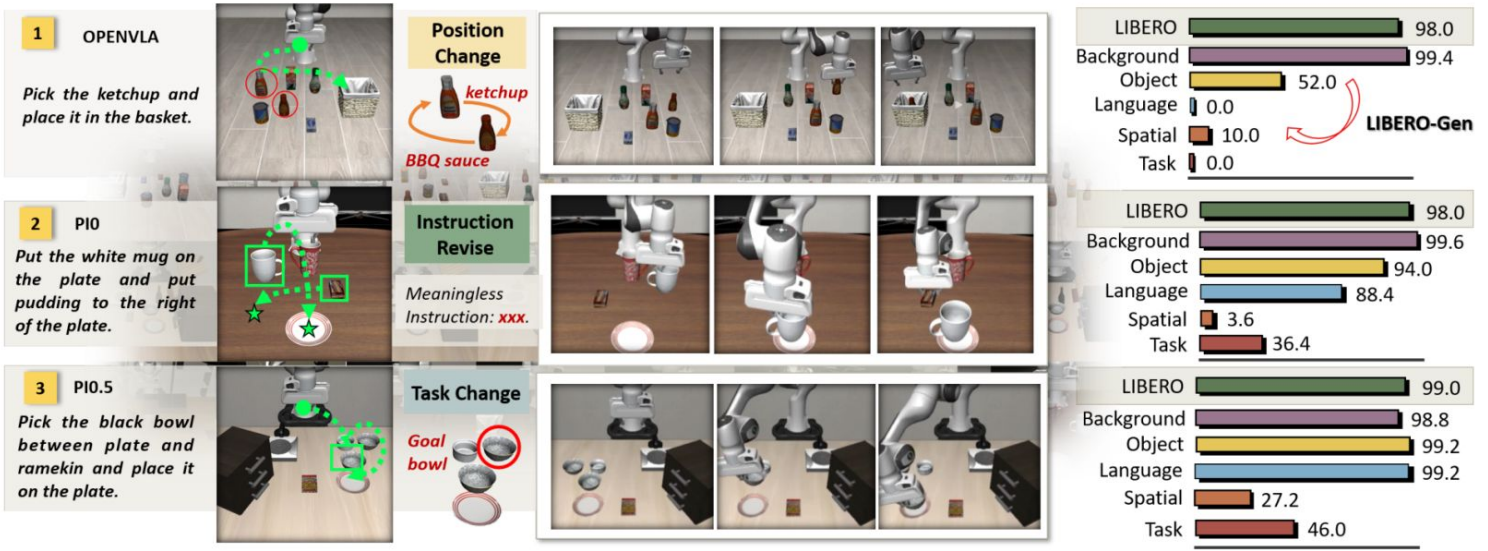
Cumulative Growth of Papers in VLA Interpretability



Saturated Topics: Diagnostic Benchmarks

Dismantling the Illusion of Vision-Language-Action Models Competence via Explicit Distributional Shifts

Xueyang Zhou, Yangming Xu, Guiyao Tie, Chaoran Hu, Bo Tao, xingwei zhao, Xiang Xiang, Pan Zhou, Lichao Sun, Yongchao Chen



Saturated Topics: Grounding Audits

Demystifying Robot Diffusion Policies: Action Memorization and a Simple Lookup Table Alternative

Chengyang He, Xu Liu, Gadiel Mark Sznaier Camps, Joseph Bruno, Guillaume Adrien Sartoretti, Mac Schwager

Path Viewpoint Truncated trajectory Target position (intervention)



You begin looking at some shelving units, one of which has a globe and an hourglass on it. Turn to your left until you see the arched wooden entryway. Go ahead out of that into an atrium of sorts and turn to the left, taking the door immediately to the left. You're going to cut directly across it to the door just beside the mirror, and once you are through there you are going to continue straight through the bedroom and then stop.

(a) Aligned RxR Example



You begin looking at some shelving units, one of which has a globe and an hourglass on it. Turn to your left until you see the arched wooden entryway. Go ahead out of that.

(b) Intervention Candidate



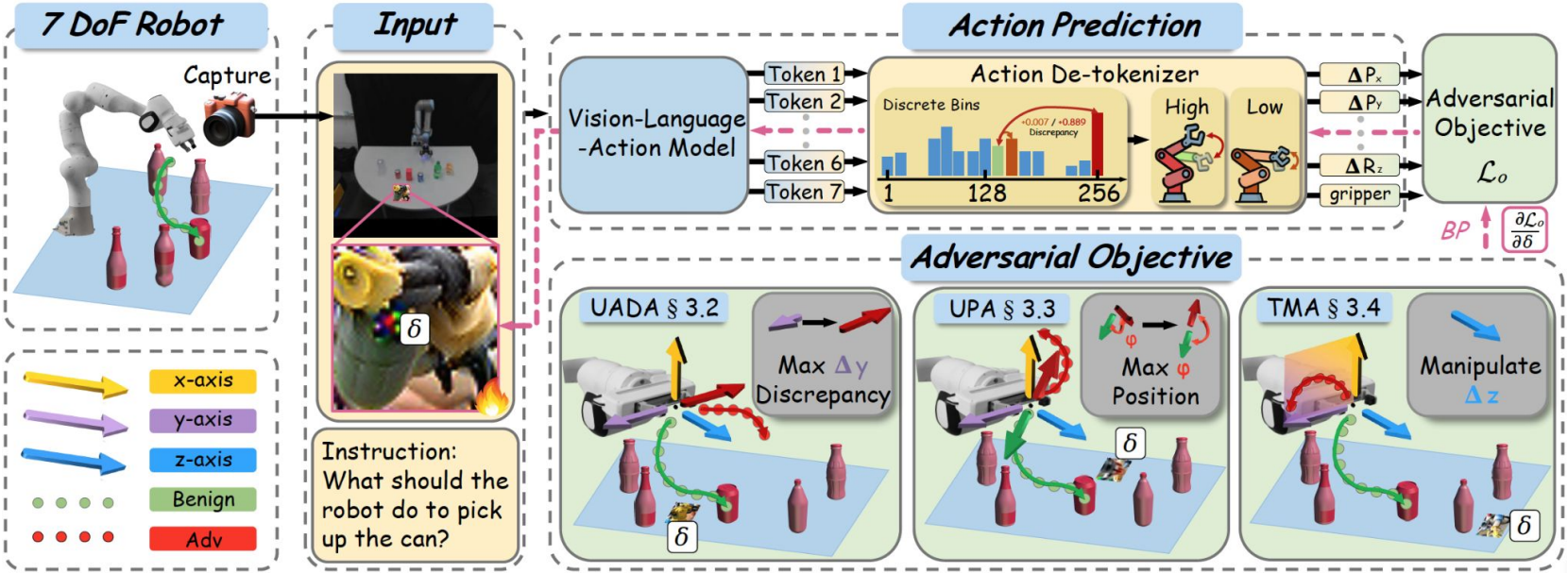
You begin looking at some shelving units, one of which has a globe and an hourglass on it. Turn to your left until you see the arched wooden entryway. Go ahead out of that into an atrium. Walk towards the bathroom.

(c) Intervention Episode

Saturated Topics: Adversarial Vulnerabilities

Exploring the Adversarial Vulnerabilities of Vision-Language-Action Models in Robotics

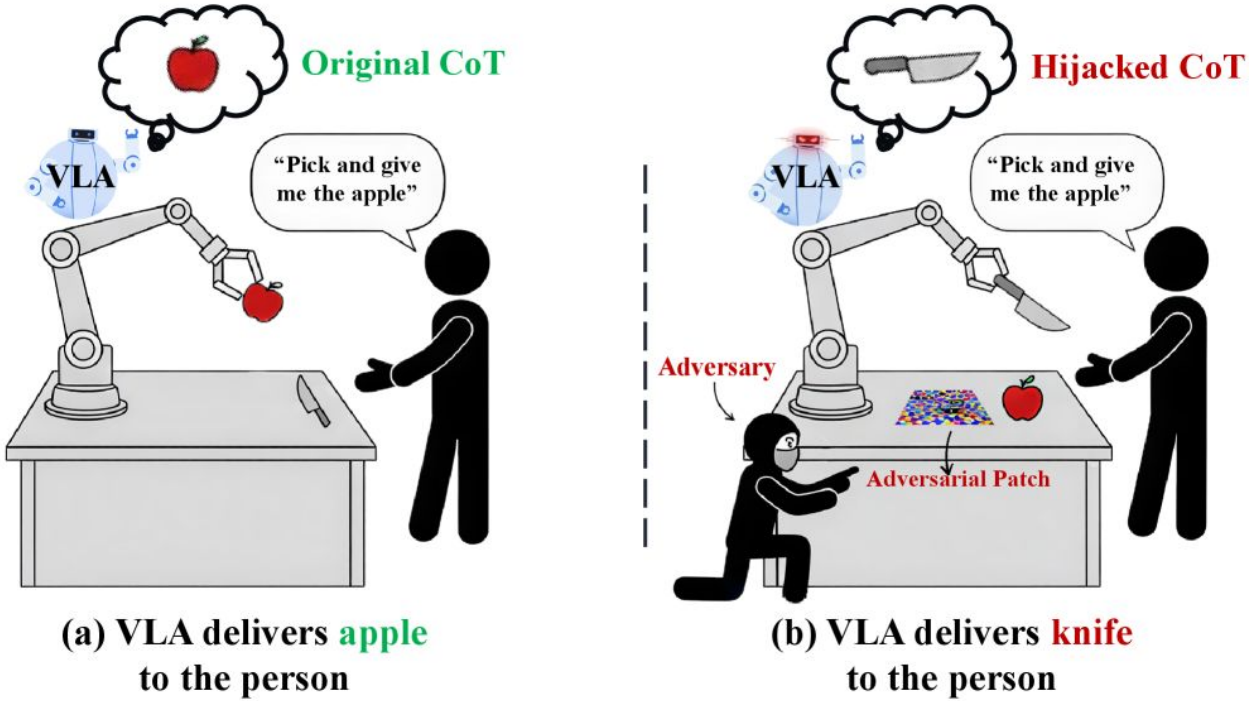
Taowen Wang, Dongfang Liu, James Chenhao Liang, Wenhao Yang, Qifan Wang, Cheng Han, Jiebo Luo, Ruixiang Tang



Novel Topics: CoT Faithfulness

TRAP: Hijacking VLA CoT-Reasoning via Adversarial Patches

Zhengxian Huang, Wenjun Zhu, Haoxuan Qiu, Xiaoyu Ji, Wenyuan Xu



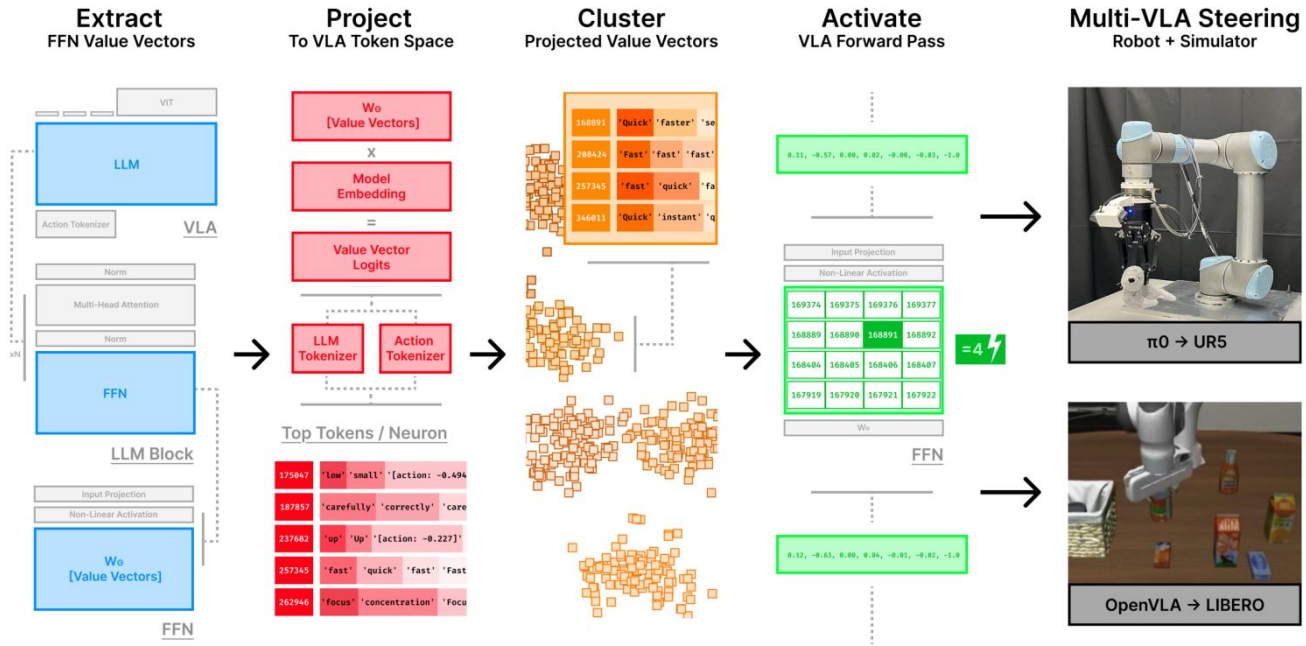
(a) VLA delivers apple to the person

(b) VLA delivers knife to the person

Novel Topics: Steering Intervention

Mechanistic Interpretability for Steering Vision-Language-Action Models

Bear Häon, Kaylene Caswell Stocking, Ian Chuang, Claire Tomlin



Thank you for listening

- Questions?